

Yi Gao

Robot Learning · Contact-Rich Manipulation · Humanoid Compliance

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Available immediately for remote research collaboration: 30 hours/week

Education

SWUFE-UD Data Science Institute

Sep 2024 – Jun 2028 (expected)

Southwestern University of Finance and Economics · Chengdu, China

Information Management and Information Systems; GPA: **87.7/100**

Four-year, China-based dual-degree program with the University of Delaware

University of Delaware degree: B.S. in Information Systems (expected)

Research Experience

SymBiosis

May 2026 – Present

Research Intern; ongoing remote collaboration · Advisor: Dr. Pengxiang Ding

SoftSONIC: Residual Learning for Humanoid Compliance

- **Independently led SoftSONIC** from method design and data generation to RL experiments and deployment validation, with research guidance from senior collaborators. Studied lightweight compliance adaptation of an existing humanoid motion prior.
- Developed a bounded **64-D latent residual** over the frozen SONIC encoder and decoder; trained it with PPO and SoftMimic-inspired compliant references. The deployed actor uses proprioceptive/action history without explicit external-force measurements.
- Trained and evaluated a **shared policy across 32 motions**, using paired perturbations and no-force tests; compared joint-space residuals, nominal-token conditioning, and reward variants to study compliance, motion preservation, and transfer.
- Implemented and validated a **50 Hz C++/TensorRT deployment** based on official SONIC. Tested standing, walking, and selected shared-policy motions on a **Unitree G1**, observing yielding under manual pulls. Larger-scale training and quantitative task evaluation are ongoing.

Lenovo Robotics Research Institute (Shanghai)

Jul 2025 – Dec 2025

Research Intern · Solution & Development Center

- Developed a context-grounded robot agent prototype for industrial patrol, integrating visual perception, location/time-specific safety rules, and action protocols into a traceable decision pipeline.
- Collected and organized **200+ real-world video clips** using a DEEPRobotics X30 quadruped; built a structured annotation and VLM evaluation workflow comparing QwenVL and GLM-family models.
- Drafted *CE4Patrol: Multi-Layer Context Reasoning for Industrial Anomaly Inspection* (manuscript; Yi Gao, et al.).

Selected Technical Project

ACT and Imitation Learning

Feb – Mar 2026

Built an ACT-style Transformer architecture with action chunking and reproduced an ACT training workflow in ManiSkill; implemented a VAE to study latent-variable learning.

Technical Skills

Programming & tools: Python, C++, PyTorch, Git, Linux, Docker, TensorRT

Simulation: Isaac Lab / Isaac Sim, MuJoCo, ManiSkill, PyBullet

Methods: PPO, residual policy learning, motion tracking, imitation learning, experimental diagnostics

Honors & Languages

Meritorious Winner, Mathematical Contest in Modeling (COMAP), 2026

Second Prize, National English Competition for College Students, 2026 · CET-4: 632; CET-6: 599